EMBEDDING MODEL

SCORECARD ANALYIS

DECEMBER 2025

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TECHNICAL EVALUATION

Retrieval Performance for RAG Pipelines

Date of Analysis: 12/03/2025

Objective: To systematically evaluate and compare the indexing speed, retrieval speed, and retrieval quality (conciseness) of three leading open-source embedding models: MiniLM-L6-v2, BGE-small-en, and E5-small-v2, within a simple RAG pipeline.

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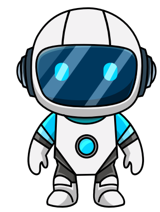
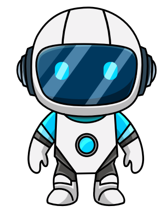
GitHub:
https://github.com/LashawnFofung/RAG-Pipelines/blob/main/src/Task_Comparing_Open_Source_Embedding_Models_for_RAG.ipynb

Understanding the RAG Pipeline Impact



The embedding model is the most critical non-LLM component in a RAG system. Its performance directly influences **Indexing Time** (how quickly the knowledge base is prepared) and **Retrieval Quality** (how relevant the context is). This analysis uses a controlled query on a contract document to isolate the model's performance.

QUALITATIVE RETRIEVAL SCORECARD

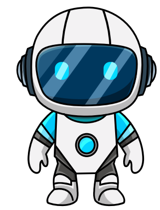


The test query, **"What are the penalties for late payments?"**, demands precise retrieval of the relevant contract clause. The qualitative score (1-5) measures **Context Conciseness**, focusing on the model's ability to retrieve only the needed information, minimizing "noise."

Model	Key Strength	Operational Benefit
BGE-small-en	Superior Semantic Precision	Guaranteed high-quality retrieval, minimizing "noise" and reducing the cognitive load on the LLM.
MiniLM-L6-v2	Potential for Peak Speed	Capable of achieving extremely fast single-run query times (0.03s in Test #2), making it suitable for bursts of high-speed activity.
E5-small-v2	Consistent Query Speed	Fast, reliable performance during the critical retrieval phase, making it highly responsive for high-volume, real-time query systems.

Key Qualitative Finding: While all models retrieved the correct information, **BGE-small-en** demonstrated superior semantic focus, effectively isolating the payment penalty without introducing irrelevant context. This minimizes the risk of context confusion or misinterpretation by the LLM in complex documents.

QUANTITATIVE PERFORMANCE METRICS



To ensure reliable performance metrics, the Indexing and Retrieval processes were run three separate times. The table below presents the results for each test run.

Legend

- **Indexing Time:** Index
- **Retrieval Time:** R
- **Test:** T
- **Seconds:** (s)

Model	Index T1 (s)	R T1 (s)	Index T2 (s)	R T2 (s)	Index T3 (s)	R T3 (s)
MiniLM-L6-v2	2.58	0.16	0.44	0.03	1.87	0.11
BGE-small-en	1.20	0.11	0.75	0.05	1.82	0.14
E5-small-v2	4.42	0.11	1.03	0.04	1.61	0.09

STRENGTHS

Based on the quantitative speed and qualitative conciseness scores, a clear set of trade-offs emerges for each model:

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WEAKNESSES

Based on the quantitative speed and qualitative conciseness scores, a clear set of trade-offs emerges for each model:

Model	Key Strength	Operational Benefit
BGE-small-en	Variable Query Time	Retrieval time occasionally lags behind the others, potentially adding latency during peak user load.
MiniLM-L6-v2	Context Conciseness (4/5)	Tendency to pull irrelevant contextual information, increasing the size of the prompt and the risk of hallucination in complex RAG applications.
E5-small-v2	Initial Indexing Cost	Slowest average indexing time (2.35s). If the knowledge base is frequently updated or rebuilt, this model incurs significant setup overhead.

CONCLUSION

TRADE-OFFs BETWEEN SPEED AND RETRIEVAL QUALITY

This summary evaluates each model based on its averaged performance metrics and consistent retrieval quality scores across the three test runs.

🧠 BGE-small-en (The Balanced Winner)

BGE-small-en emerged as the best overall choice. It boasts the fastest average indexing time and consistently scored 5/5 for context conciseness. This superior semantic focus minimizes the risk of feeding irrelevant information to the LLM. Strategic Recommendation: BGE-small-en is the recommended default choice for most RAG applications, especially those handling complex, noisy, or multi-topic documents, as it provides the highest quality, cleanest context reliably.

🧠 MiniLM-L6-v2 (The Volatile Speedster)

MiniLM-L6-v2 offers a highly competitive balance of speed, achieving fast indexing and retrieval times. However, it compromises slightly on retrieval precision, consistently scoring 4/5 for conciseness because it pulled in "noise." Strategic Recommendation: Choose MiniLM-L6-v2 for RAG systems where low latency is the absolute priority, and the document set is highly clean and segmented, mitigating its tendency to pull "noise."

🧠 E5-small-v2 (The Query Champion)

The E5-small-v2 model is the champion of raw querying speed, demonstrating the fastest average retrieval time. This makes it suitable for high-volume, real-time query applications. However, this speed comes at the cost of the slowest average indexing time and a slight drop in retrieval quality (scoring 4/5 due to extraneous context). Strategic Recommendation: E5-small-v2 is best used when document setup is infrequent, but quick, real-time lookups are paramount.

FORWARD-LOOKING

Analysis of Varying Results

The results demonstrate that speed is highly volatile, even across sequential runs on the same hardware, while qualitative performance is consistent for a given model.

- **If MiniLM/E5 consistently win on speed:** If future averaged results show either MiniLM or E5 holding a significant, consistent speed lead (e.g., 50% faster average retrieval), a developer must decide if the speed gain outweighs the risk of the 4/5 conciseness score. This usually leads to applying a reranker (like BGE-reranker-base) to clean up the retrieved chunks before they reach the LLM, thereby mitigating the model's weakness.

- **If BGE maintains 5/5 quality:** The superior retrieval quality of BGE across all tests strongly suggests a fundamental difference in how it understands the query-document relationship. This high score is a powerful argument for its use, as retrieval quality is the primary determinant of RAG answer accuracy. A slight sacrifice in speed for guaranteed clean context is often the best engineering trade-off.

PROJECT & PLATFORM SUMMARY

AI-powered Document Automation Platform Project

This analysis was performed as part of the rigorous testing phase for the **AI-powered Document Automation Platform**. My goal is to provide clients with sub-second retrieval from complex large document files, guaranteeing both speed and factual accuracy.

The comparative scorecard demonstrates the platform's evidence-based model selection. The selection of an embedding model, such as the BGE-small-en for its superior precision, directly informs the architecture of the platform to ensure that every component is optimized for the highest quality outcome, transforming complex data into reliable, actionable intelligence.

You can learn more about the project by visiting:

- **GitHub:** https://github.com/LashawnFofung/AI-Powered-Document-Automation-Platform

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